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Patent Public Search | Text View

United States Patent Application Publication

20250276118

Kind Code

A1

Publication Date

September 04, 2025

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RENAL FAILURE THERAPY METHOD FOR ELECTRICALLY SAFE TREATMENT

Abstract

A method for powering an external device through a renal failure therapy system is disclosed herein. The renal failure therapy system includes a dialyzer, a blood circuit in fluid communication with the dialyzer, a dialysis fluid circuit in fluid communication with the dialyzer, a housing supporting the dialyzer, the blood circuit, and the dialysis fluid circuit, and at least one electrical socket held by the housing. The method comprises providing, through the at least one electrical socket, a voltage output to a respective external electrical device for powering or charging the respective external electrical device. The at least one electrical socket includes electrical insulation for protecting a patient while powering or charging the respective external electrical device.

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Family ID: 57133220

Appl. No.: 19/211697

Filed: May 19, 2025

Foreign Application Priority Data

SE	1551325-2	Oct. 14, 2015
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Related U.S. Application Data

parent US continuation 15766185 20180405 parent-grant-document US 10926018 WO

continuation PCT/EP2016/074710 20161014 child US 17178005

parent US division 17178005 20210217 parent-grant-document US 12303629 child US 19211697

Publication Classification

Int. Cl.: A61M1/34 (20060101); **A61M1/16** (20060101); **A61M1/36** (20060101)

U.S. Cl.:

CPC **A61M1/3413** (20130101); **A61M1/16** (20130101); **A61M1/1621** (20140204);
 A61M1/34 (20130101); **A61M1/3621** (20130101); **A61M1/3624** (20130101);
 A61M1/367 (20130101); A61M2205/3331 (20130101); A61M2205/3351 (20130101);
 A61M2205/3368 (20130101); A61M2205/502 (20130101)

Background/Summary

PRIORITY CLAIM [0001] The present application is a divisional application of U.S. application Ser. No. 17/178,005, filed Feb. 17, 2021, which is a continuation application of U.S. application Ser. No. 15/766,185, filed Apr. 5, 2018, now U.S. Pat. No. 10,926,018, which is a National Phase of International Application No. PCT/EP2016/074710, filed Oct. 14, 2016, which claims priority to Swedish Application No. 1551325-2, filed Oct. 14, 2015. The entire contents of each are incorporated herein by reference and relied upon.

BACKGROUND

[0002] The present disclosure relates generally to medical systems. More specifically, the present disclosure relates to medical systems that allow a patient to safely power an external electronic device during treatment.

[0003] Hemodialysis (“HD”) in general uses diffusion to remove waste products from a patient's blood. A diffusive gradient that occurs across the semi-permeable dialyzer between the blood and an electrolyte solution called dialysis fluid causes diffusion. Hemofiltration (“HF”) is an alternative renal replacement therapy that relies on a convective transport of toxins from the patient's blood. This therapy is accomplished by adding substitution or replacement fluid to the extracorporeal circuit during treatment (typically ten to ninety liters of such fluid). The substitution fluid and the fluid accumulated by the patient in between treatments is ultrafiltered over the course of the HF treatment, providing a convective transport mechanism, which is particularly beneficial in removing middle and large molecules (in hemodialysis there is a small amount of waste removed along with the fluid gained between dialysis sessions, however, the solute drag from the removal of that ultrafiltrate is typically not enough to provide convective clearance).

[0004] Hemodiafiltration (“HDF”) is a treatment modality that combines convective and diffusive clearances. HDF flows dialysis fluid through a dialyzer, similar to standard hemodialysis, providing diffusive clearance. In addition, substitution solution is provided directly to the extracorporeal circuit, providing convective clearance. These modalities are administered by a dialysis machine. The machines may be provided in a center or in a patient's home. Dialysis machines provided in a center are used multiple times a day for multiple patients and therefore must be cleaned between treatments. Dialysis machines use multiple components, including electrical components.

[0005] Outside electrical devices have the potential to expose people to the risk of spurious electric currents. In the case of medical electrical equipment (and dialysis machines in particular), the risk is potentially greater how depending upon the equipment is connected to the patient. Patients connected to present dialysis machines may be poorly safeguarded against leakage current due to contact with external electrical equipment, such as bed lamps, electrically adjustable beds or treatment chairs, lap tops, phones connected to chargers, and other electrical equipment that is in turn connected to a mains electrical power source. Poor quality chargers are especially dangerous as their electrical insulation may break down, leading to fault currents that may run to the patient. It is accordingly advisable for dialysis clinics to ask that patients during treatment not plug their

electronic devices into an external power source, which may be inconvenient for patients undergoing a treatment that may last for hours.

[0006] It is accordingly desirable to provide a system that allows a patient to use and power an external electronic device during treatment safely.

SUMMARY

[0007] The present disclosure provides a renal failure therapy system and method that performs hemodialysis (“HD”), hemofiltration (“HF”) and hemodiafiltration (“HDF”). Accordingly, “renal failure therapy” as used herein is meant to include any one, or more, or all of HD, HF and/or HDF.

[0008] The renal failure therapy system and method of the present disclosure includes a machine providing at least one electrically insulated electrical socket, which enables a patient undergoing treatment to power an external device, such as a smartphone, personal computer, reading lamp and the like. The goal of the electrical insulation is to provide an extra galvanic separation between the patient and the machine to minimize leakage or fault currents. For AC current sockets, the electrical insulation may include a transformer that provides galvanic isolation forming an open circuit, which prevents propagation of fault currents. The transformer has an input coil separated physically from an output coil. The input coil magnetically induces a desired AC voltage in the output coil. The desired AC output voltage charges or powers the user's electrical device, such as a computer.

[0009] For DC sockets, the electrical insulation may take multiple forms. One form of DC insulation uses DC to AC conversion. The DC input voltage is converted to an AC input voltage, which then powers an input coil. The AC input coil magnetically induces an AC output voltage in an output coil as described above. Here, however, the AC output voltage is converted into a desired DC voltage, typically 5 VDC, 12 VDC or 24 VDC, for powering and/or charging the user's electrical device. Other DC insulation providing galvanic protection for low voltages may include optical means.

[0010] The machine may have multiple electrically insulated sockets, for example, providing an array of desirable output voltages of 5 VDC, 12 VDC, 24 VDC, 120 VAC, and/or 220 VAC. It is contemplated to provide banks of electrically insulated sockets of the same output voltage, for example, a bank of sockets providing output voltages of 5 VDC. Where coiled transformers are used, the banks may have a single primary coil for multiple secondary coils, or a single primary coil for each secondary coil.

[0011] The electrically insulated electrical sockets may be located on the front of the renal failure therapy machine, or at another location on the machine that is readily accessible to the patient and/or nurse. It is ensured that the electrically insulated electrical sockets are well made, so that they provide robust electrical insulation. Indeed, the sockets may provide two layers of electrical insulation (or double insulation). In this manner, it is assured that the patient may safely plug his or her electrical device into the renal failure therapy machine during treatment for operation and/or charging.

[0012] As discussed, the electrically insulated sockets allow the user to safely power their external electrical devices during treatment. To make treatment even safer, it is contemplated to combine the electrically insulated sockets with other electrical insulation, which protects the patient electrically towards a fluid path in the renal failure therapy machine. That is, one may think of the patient as being placed electrically between the external electrical device connected to the clinic's mains power system and electrical earth via a path that runs through the blood lines, the patient, the conductive dialysis fluid lines in the machine, and outside the machine through an external used dialysate drain line. The electrically insulated sockets place a layer (or double layer) of electrical protection between the patient and the clinic's mains power system. It is also contemplated to place a layer of electrical protection between the patient and electrical earth via the conductive fluidic pathway through the renal failure therapy machine.

[0013] In one embodiment, the layer of electrical protection between the patient and electrical earth includes an electrically floating fluid pathway. Generally, an electrically floating fluid pathway is

one that is not connected to electrical earth. As used herein, electrically floating fluid pathway in one embodiment means instead that there is no pathway to electrical earth within the blood lines, dialysis fluid lines either inside the machine or outside the machine towards the dialyzer, concentrate lines, or even the water lines (e.g., if the water is non-deionized). That is, electrically floating fluid pathway may mean a fluid pathway which, when carrying an electrically conductive fluid therein, would itself render the conductive fluid electrically floating relative to an electrical potential, such as electrical earth, provided to the dialysis machine through the mains and/or through earthed parts connected to the dialysis machine (e.g. drain and external water lines). The electrically floating fluid pathway may include the entire or one or several portion(s) of the blood lines, (fresh and/or used) dialysis fluid lines, concentrate lines, and/or water lines as well as components, such as sensors and pumps, connected to the above mention fluid lines. The only pathway to electrical earth is via the used dialysis fluid traveling outside the machine through an external drain line to electrical earth, for example, at the clinic's house drain. Making any fault voltages generated at the patient travel all the way to electrical earth at the house drain increases the naturally occurring impedances within the fluid lines that the fault voltage sees, thereby minimizing the current generated by the fault voltage.

[0014] There are a number of structural modifications made to allow the machine to operate with an electrically floating fluid pathway. First, dialysis machines typically intentionally connect various flow components to protective earth. For example, sensing equipment is typically connected to a signal ground to divert fault currents away from the sensing equipment to prevent false readings. The signal ground may be connected to protected earth in the dialysis machine directly or indirectly via the dialysis fluid. The electrically floating pathway of the present disclosure cannot exist with such connections to protective earth. Instead, the sensitive equipment is here provided with electrical bypassing, which bypasses fault currents around the equipment, through the electrically floating fluid pathway and through an external used dialysate drain line, to electrical earth at the house drain.

[0015] Second, certain flow components, such as the sensing equipment, have probes or other conductive structures that contact the medical fluid, such as dialysis fluid. Any such component becomes a potential pathway to electrical earth due to its wiring. In the present disclosure, the wiring of any such component is electrically isolated with a single or double layer of electrical insulation in the same manner as the electrically insulated electrical sockets discussed herein.

[0016] The above-described combination of electrical insulation prevents or reduces fault voltages generated due to the powering of a patient's external personal electrical equipment (a notoriously large source of fault voltages) from occurring. The combination also forces any fault voltages that don't occur at the patient or somewhere within the machine to dissipate along a relatively high impedance fluid path, through the electrically floating fluid pathway and through an external used dialysate drain line, to the electrical earth at the house drain.

[0017] In light of the technical features set forth herein, and without limitation, in a first aspect, a renal failure therapy system includes: a dialyzer; a blood circuit in fluid communication with the dialyzer; a dialysis fluid circuit in fluid communication with the dialyzer; a housing supporting the dialyzer, the blood circuit and the dialysis fluid circuit; and at least one electrical socket held by the housing, the electrical socket providing a voltage output dedicated to a particular voltage type of external electrical device for powering or charging the external electrical device, the at least one electrical socket including electrical insulation for protecting a patient while powering the external electrical device.

[0018] In a second aspect, which may be used in combination with any other aspect described herein unless specified otherwise, the voltage output dedicated to a particular voltage type of external electrical device is a USB voltage output.

[0019] In a third aspect, which may be used in combination with any other aspect described herein unless specified otherwise, the voltage output dedicated to a particular voltage type of personal

external device is a 5, 10 or 24 VDC output.

[0020] In a fourth aspect, which may be used in combination with any other aspect described herein unless specified otherwise, the voltage output dedicated to a particular voltage type of personal external device is a 120 or 240 VAC output.

[0021] In a fifth aspect, which may be used in combination with any other aspect described herein unless specified otherwise, the electrical insulation of the at least one electrical socket includes a transformer.

[0022] In a sixth aspect, which may be used in combination with any other aspect described herein unless specified otherwise, the electrical insulation of the at least one electrically insulated electrical socket includes a DC to DC converter.

[0023] In a seventh aspect, which may be used with the sixth aspect in combination with any other aspect described herein unless specified otherwise, the DC to DC converter includes an input conductively insulated from an output.

[0024] In an eighth aspect, which may be used in combination with any other aspect described herein unless specified otherwise, the electrical insulation of the at least one electrical socket includes a transformer in combination with (i) an AC to DC converter or (ii) a DC to AC converter and an AC to DC converter.

[0025] In a ninth aspect, which may be used in combination with any other aspect described herein unless specified otherwise, the renal failure therapy system includes a bank of electrical sockets providing a first voltage output dedicated to a same particular voltage type of the external electrical device.

[0026] In a tenth aspect, which may be used with the ninth aspect in combination with any other aspect described herein unless specified otherwise, the bank is a first bank, and which includes a second bank of electrical sockets providing a second voltage output dedicated to a same particular voltage type of second external electrical device, the second voltage different from the first voltage.

[0027] In an eleventh ninth aspect, which may be used in combination with any other aspect described herein unless specified otherwise, the renal failure therapy system includes an electrically floating fluid pathway provided in at least a portion of the blood circuit and at least a portion of the dialysis fluid circuit, wherein the only electrical path to ground is via used dialysis fluid traveling through the machine to electrical earth.

[0028] In a twelfth aspect, which may be used with the eleventh aspect in combination with any other aspect described herein unless specified otherwise, at least one electrically sensitive component in the at least portion of the dialysis fluid circuit of the electrically floating fluid pathway is electrically bypassed.

[0029] In a thirteenth aspect, which may be used with the twelfth in combination with any other aspect described herein unless specified otherwise, the at least one electrically sensitive component is electrically insulated from its power and signal input wires.

[0030] In a fourteenth aspect, which may be used with the eleventh aspect in combination with any other aspect described herein unless specified otherwise, the renal failure therapy system includes an external drain line and a drain having an electrical earth, and wherein the electrically floating fluid pathway leads to the external drain line, which leads to the drain.

[0031] In a fifteenth aspect, which may be used in combination with any other aspect described herein unless specified otherwise, a renal failure therapy machine for providing treatment to a patient using an external electrical device includes: a dialyzer; a blood circuit in fluid communication with the dialyzer; a dialysis fluid circuit in fluid communication with the dialyzer; a housing supporting the dialyzer, the blood circuit and the dialysis fluid circuit; and an electrical socket held by the housing, the electrical socket providing a voltage output dedicated to a voltage of the external electrical device for powering or charging the external electrical device, the electrical socket including electrical insulation for protecting a patient while powering the external electrical device.

[0032] In a sixteenth aspect, which may be used with the fifteenth aspect in combination with any other aspect described herein unless specified otherwise, there is no protective earth connection within or on the housing to any fluid flowing within the dialysis fluid circuit.

[0033] In a seventeenth aspect, which may be used with the fifteenth aspect in combination with any other aspect described herein unless specified otherwise, there is no protective earth connection within or on the housing to any fluid flowing within the blood circuit or dialyzer.

[0034] In an eighteenth aspect, which may be used with the fifteenth aspect in combination with any other aspect described herein unless specified otherwise, the electrical socket is a first socket and the voltage output is a first voltage output, and wherein the housing supports a second electrical socket providing a second voltage output different than the first voltage output.

[0035] In a nineteenth aspect, which may be used with any other aspect described herein unless specified otherwise, the electrical insulation is or includes double electrical insulation.

[0036] In a twentieth aspect, any of the features, functionality and alternatives described in connection with any one or more of FIGS. 1 to 7 may be combined with any of the features, functionality and alternatives described in connection with any of the other one or more of FIGS. 1 to 7.

[0037] It is therefore an advantage of the present disclosure to provide a hemodialysis, hemofiltration or hemodiafiltration system and method having electrically insulated electrical sockets.

[0038] It is another advantage of the present disclosure to provide a hemodialysis, hemofiltration or hemodiafiltration system and method having electrically insulated electrical sockets in combination with electrically insulated input lines to the socket to provide double insulated electrical sockets.

[0039] It is another advantage of the present disclosure to provide a hemodialysis, hemofiltration or hemodiafiltration system and method having electrical sockets that enable a patient to safely power and/or charge an external electrical device during treatment.

[0040] It is a further advantage of the present disclosure to provide a hemodialysis, hemofiltration or hemodiafiltration system and method having electrically insulated electrical sockets provided in combination with fluid lines that are additionally electrically insulated.

[0041] It is yet another advantage of the present disclosure to provide a hemodialysis, hemofiltration or hemodiafiltration system and method having electrically insulated electrical sockets provided in combination with fluid lines that are electrically floating within the blood tubing and the renal failure treatment machine.

[0042] It is yet a further advantage of the present disclosure to provide a hemodialysis, hemofiltration or hemodiafiltration system and method having electrical sockets that enable a patient to safely power and/or charge an external electrical device during treatment in combination with electrical insulation that mitigates the effects of fault voltages that may still be generated at the patient, in the blood lines, or in the dialysis fluid lines.

[0043] The advantages discussed herein may be found in one, or some, and perhaps not all of the embodiments disclosed herein. Additional features and advantages of the present invention are described in, and will be apparent from, the following Detailed Description of the Invention and the figures.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0044] FIG. 1 illustrates one example of a potential electrical insulation problem with existing renal failure therapy machines.

[0045] FIG. 2 is a side elevation view of one embodiment of an electrically insulated renal failure therapy machine of the present disclosure.

[0046] FIGS. 3A and 3B are exploded side and front elevation views, respectively, illustrating one example of the insulated electrical sockets of the present disclosure.

[0047] FIG. 4 is a side exploded view illustrating another example of the insulated electrical sockets of the present disclosure.

[0048] FIG. 5 is a schematic view illustrating various types of electrical conversion and isolation suitable for use with the insulated electrical sockets of the present disclosure.

[0049] FIG. 6 illustrates one embodiment of a dialysis fluid circuit forming part of an electrically floating fluid pathway, which may be used with the insulated electrical sockets of the present disclosure.

[0050] FIG. 7 illustrates one embodiment of a blood fluid circuit forming part of an electrically floating fluid pathway, which may be used with the insulated electrical sockets of the present disclosure.

DETAILED DESCRIPTION

[0051] Referring now to the drawings and in particular to FIG. 1, a known dialysis machine **8** connected to a patient **116** illustrates a potential electrical insulation problem. Known machine **8** is connected within its housing to machine protective earth **27** in multiple places to protect various components, such as conductivity sensors, which may read improperly when subjected to currents due to fault voltages. The system's protective earth connections take the fault currents instead to protective earth **27**, protecting the sensing equipment. Machine **8** is also connected outside its housing to electrical earth **28**.

[0052] Protective earth **27** for known machine **8** is also intended to ensure that the patient is safe from an internal electrical failure occurring within machine **8**. However, an electrical risk to patient **116** is created because the patient is electrically connected via conductive dialysis fluid to protective earth **27**. If patient **116** touches a faulty piece of electrical equipment having an electrical potential relative to protective earth, a potentially dangerous current may flow from the faulty equipment through patient **116**, to protective earth **27**.

[0053] Patient **116** in the illustrated embodiment is handling an external electrical device **200**, which may be any type of electrical device, such as a reading lamp, heating pad, etc. In many cases, external electrical device **200** will be a personal communication/computing device ("PCD"), such as a smart phone, tablet, or personal computer. In any case, the external electrical device **200** is connected to mains power in FIG. 1.

[0054] Many PCD's are provided with cheaply made chargers, which are prone to failure. When the chargers fail, a very real potential for a fault voltage is created. When a fault voltage is created, if there is a path to electrical earth **28**, a fault current will be generated. The system's connections to protective earth **27** within known machine **8**, which provide low impedance paths to protect the sensing equipment, also provide a low impedance path for the fault current.

[0055] In many instances, needle access to patient **116** is into a vein in one of the patient's arms. FIG. 1 illustrates an alternative needle access established in which a central venous catheter ("CVC") is inserted into vein very close to the patient's heart. In this scenario, fault voltages, such as those just described, present a heightened risk due to the proximity to the patient's heart.

[0056] It is advisable for dialysis clinics to prevent the patient from powering an external electrical device **200** using the clinic's mains power during treatment. It is difficult to ensure however that every clinic will follow this procedure for every treatment. Certain patients, perhaps not understanding the risk, may disobey the rules and plug their device **200** into the mains power system when left unattended. Moreover, chronic kidney failure treatments may last hours, leading to an ever-increasing desire for the patients to want to power their PCD's **200**.

[0057] Referring now to FIG. 2, system **10** having renal failure therapy machine **12** provides a solution to the above described problem. Machine **12** provides a renal failure therapy to patient **116**. The components of machine **12** are discussed in detail below. For now, it is enough to know that machine **12** may include internal components that prepare a dialysis fluid, and pump such

dialysis fluid along a fresh dialysis fluid line **76** to a dialyzer **102**. The internal components of machine **12** also pump used dialysis fluid from dialyzer **102** along a used dialysis fluid line **56** to an external drain line **57** and to drain **60**.

[0058] Machine **12** also includes a blood pump that pumps blood from patient **116** along an arterial line **106** and pushes the blood through dialyzer **102** and venous line **108**, back to patient **116**. The blood pump **120**, dialyzer **102**, arterial line **106** and venous line **108** are typically located at the front **12a** of machine **12**. Machine front **12a** is accordingly generally open and accessible to the nurse and patient and provides a convenient location **160** for one or more electrically insulated electrical socket **170**. In alternative embodiments, insulated socket location **160** is provided on any one or more of machine top **12b**, machine sides **12c** and **12d** (**12d** not viewable in FIG. 2), or machine backside **12e**.

[0059] In the illustrated embodiment, location **160** includes differently rated insulated electrical sockets **170a**, **170b**, and **170c** (referred to herein collectively as sockets **170** and generally, individually as socket **170**). Sockets **170** may for example be dedicated to Universal Serial Bus (“USB”) charging **170a**, 12 VDC or 24 VDC charging **170b**, or 120 VAC or 220 VAC charging **170c**. Each socket **170a**, **170b**, and **170c** is accordingly dedicated to a particular voltage type of external electrical device **200**. The side section view of FIG. 2 is only able to show a single (“USB”) socket **170a**, a single 12 VDC or 24 VDC socket **170b**, and a single 120 VAC or 220 VAC socket **170c**. It should be appreciated that there may be a bank **162** of multiple (“USB”) sockets **170a**, a bank **164** of multiple 12 VDC or 24 VDC sockets **170b**, and a bank **166** of 120 VAC or 220 VAC sockets **170c**.

[0060] Referring now to FIG. 3A, an exploded view of the socket bank location **160** of FIG. 2 is illustrated. For ease of illustration, each of USB sockets **170a**, 12 VDC or 24 VDC sockets **170b**, and 120 VAC or 220 VAC sockets **170c** is illustrated as having a transformer **172** having a primary coil **174** and a secondary coil **176**. Each primary coil **174** is separated from a secondary coil **176** by an isolation barrier **178**, which may be air gap or gap filled with electrically resistive potting material. Isolation barrier **178** provides a conductive break, which conductively isolates components “downstream” from secondary coil **176**, including patient **116**, from any electrically connected component upstream of primary coil **174**, including mains voltage illustrated in FIG. 2. Primary coil **174** and secondary coil **176** are magnetically coupled, such that primary coil **174** induces a desired AC voltage in secondary coil **176**.

[0061] Primary coil **174** and secondary coil **176** of transformer **172** is one preferred way of isolating 120 VAC or 220 VAC sockets **170c** and for providing the desired AC voltage for the power sockets. Patient **116** may use 120 VAC or 220 VAC sockets **170c** for example to power a personal computer **200** using the patient's existing computer power cord. Computer power cords typically provide their own transformers, which output a desired voltage to the computer **200**. The transformers may provide their own electrical insulation, which in certain instances is unreliable. 120 VAC or 220 VAC sockets **170c** provide reliable electrical insulation, which protects the patient even if the insulation of the computer cord transformer is faulty.

[0062] DC sockets **170a** and **170b** and their corresponding electrical isolation may be provided in a variety of different ways. In one embodiment the input DC voltage is converted to an input AC voltage using a DC to AC converter (illustrated in FIG. 5). The input AC voltage is conductively isolated from an AC output voltage via a transformer **172** and its isolation barrier **178** as illustrated in FIG. 3A, wherein primary coil **174** induces a desired AC output voltage in secondary coil **176**. An AC to DC converter (illustrated in FIG. 5) is provided on the output side of transformer **172** to convert the induced AC output voltage to a desired DC output, e.g., 5 VDC, 12 VDC or 24 VDC.

[0063] FIG. 3B shows a front view of FIG. 3A, better illustrating banks **162**, **164**, and **166** of electrically insulated sockets **170a** to **170c**. As illustrated, for example, there may be nine total sockets, with a bank **162** of three USB sockets **170a**, a bank **164** of multiple 12 VDC or 24 VDC sockets **170b**, and a bank **166** of 120 VAC or 220 VAC sockets **170c**. Or perhaps there are multiple

USB sockets **170a**, multiple 12 VDC or 24 VDC sockets **170b**, but only a single 120 VAC or 220 VAC socket. The combination of sockets may be chosen as desired to meet the most likely demand. [0064] Referring now to FIG. 4, an alternative embodiment for providing DC sockets **170a** and **170b** is illustrated. Here, a DC to DC converter **180** receives system DC power, e.g., 24 VDC, via system power and power ground wires **182** and **184**, respectively. An input **180a** of DC to DC converter **180** receives DC input power (e.g., 24 VDC) from system power and power ground wires **182** and **184**. Input **180a** charges output **180b** while being conductively isolated from output **180b**. DC to DC converter **180** may perform the isolated charging in different ways as discussed below. [0065] In any of the embodiments discussed above, transformer **172** may include a dedicated primary coil **174** for each output coil **176**. Alternatively, there may be a single primary coil **174** for multiple output coils **176**, where the output coils **176** may be of the same or different output voltage (e.g., different output coils may have different winding sizes and/or turns ratios to produce a different output voltage). For DC to DC conversion, a DC input voltage may be converted to an AC input voltage using a DC to AC converter, where the AC input voltage powers a common coil for multiple output coils, which each produce an AC output voltage. To this end, transformer **172** can have multiple windings or windings with several connection points (taps) on both the primary and secondary or output sides of transformer **172**. An AC to DC converter may also be provided for each AC output voltage to produce multiple DC output voltages each having a desired DC output level. For desired DC outputs, it is also contemplated to begin with an AC input voltage, induce an AC output voltage on output coil **176**, and convert the AC output voltage to a desired one or more DC output voltage.

[0066] FIG. 5 summarizes that machine **12** of system **10** may provide electrical insulation for any combination of electrically insulated sockets **170a** to **170c** via any combination of (i) straight AC to AC transformers **172**, (ii) straight DC to DC converters **180**, (iii) AC to AC transformers **172** in combination with AC to DC conversion **192**, (iv) AC to AC transformers **172** in combination with DC to AC conversion **194** and AC to DC conversion **192**, (v) AC to DC conversion **192** in combination with DC to DC conversion **180**, and (vi) DC to AC conversion **194** in combination with AC to DC conversion **192** without a transformer **172**. Other options providing galvanic separation for the patient include capacitive coupling for AC power, Hall effect sensors, and optical devices for DC power.

[0067] FIGS. 3A, 3B and 4 show in detail a first layer of electrical insulation between patient **116** and mains power provided by electrically insulated sockets **170a**, **170b** and **170c**. FIG. 2 illustrates a second layer of electrical insulation between patient **116** and mains power, namely, a mains transformer **152** having a mains input coil **154** and a mains output coil **156** separated by an isolation barrier **158**. Mains transformer **152** is located between mains power and the input or inputs to electrically insulated sockets **170a**, **170b** and **170c**. Isolation barrier **158** prevents the conducting of electricity, while input coil **154** induces a desired voltage in output coil **156**, as has been described above.

[0068] FIG. 2 illustrates that output coil **156** of mains transformer **152** powers the input sides of each of the transformers of electrically insulated sockets **170a**, **170b** and **170c**. This may occur if, for example, the output AC voltages at output coils **176** of the transformers **172** of DC sockets **170b** and **170c** are converted within the sockets using AC to DC converters **152** (FIG. 5) to the desired DC voltages. Alternatively, output coil **156** of mains transformer **152** is split into an AC power circuit and one or more DC power circuit (not illustrated) using one or more AC to DC converter. Here, the output of the AC power circuit powers the inputs of electrically insulated AC socket **170a**, while the one or more DC power circuit powers the inputs of electrically insulated DC sockets **170b** and **170c**, e.g., using isolated DC to DC.

[0069] FIG. 2 illustrates the double electrical insulation between external electrical device **200** and mains power as has just been described. Again, the double insulation helps to prevent the patient from being subjected to stray or fault voltages due for example to a faulty external electrical device

200. FIG. 2 also illustrates that system **10** provides additional electrical insulation, which helps to mitigate the effects of a stray or fault voltage that does occur due for example to a faulty external electrical device **200**. As illustrated in FIG. 2, machine impedance $M_{sub}imp$ is made to be as large as possible so as to reduce the current resulting from a fault or stray voltage as much as possible. To do so, no connection is made between (ai) the blood in blood set, (aii) the dialysis fluid in fresh or used dialysis fluid lines, (aiii) the concentrates in concentrate lines, and (aiv) water in water lines (which may not be properly deionized, each of (ai) to (aiv) is considered to be at least somewhat conductive) and (b) machine or electrical earth **28**. As illustrated in FIG. 1, prior art machines are typically connected to protective earth **27** at multiple places, for example, in the used dialysis line just downstream of the dialyzer. Configured as described above, there is no physical connection to electrical earth **28** within the fluid pathways of system **10**, such that the pathways within the machine are said to be electrically floating, causing any fault or stray currents to flow through the increased machine impedance $M_{sub}imp$ and through an external drain line **57** to electrical earth **28** at drain **60** located outside of machine **12**, thereby reducing the fault or stray current.

[0070] FIGS. 6 and 7 illustrate an embodiment of an electrically floating fluid pathway **140** of system **10**. System **10** in FIG. 6 includes a machine **12** having an enclosure or housing. The housing may provide the insulated electrical socket **170** and described above. The housing of machine **12** holds the contents of a dialysis fluid or dialysis fluid circuit **30** described in detail below. The housing or machine **12** also supports a user interface **14**, which allows a nurse or other operator to interact with system **10**. User interface **14** may have a monitor screen operable with a touch screen overlay, electromechanical buttons, e.g., membrane switches, or a combination of both. User interface **14** is in electrical communication with at least one processor **16** and at least one memory **18**. At least one processor **16** and at least one memory **18** also electronically interact with, and where appropriate, control the pumps, valves and sensors described herein, e.g., those of dialysate circuit **30**. At least one processor **16** and at least one memory **18** are referred to collectively herein as a logic implementer **20**. The dashed lines extending from logic implementer **20** lead to pumps, valves, sensors, the heater and other electrical equipment, as indicated by like dashed lines leading from the pumps, valves, sensors, heater, etc.

[0071] Dialysis fluid circuit **30** includes a purified water line **32**, an A-concentrate line **34** and a bicarbonate B-concentrate line **36**. Purified water line **32** receives purified water from a purified water device or source **22**. The water may be purified using any one or more process, such as, reverse osmosis, carbon filtering, ultraviolet radiation, electrodeionization (“EDI”), and/or ultrafiltering.

[0072] An A-concentrate pump **38**, such as a peristaltic or piston pump, pumps A-concentrate from an A-concentrate source **24** into purified water line **32** via A-concentrate line **34**. Conductivity cell **40** measures the conductive effect of the A-concentrate on the purified water, sends a signal to logic implementer **20**, which uses the signal to properly proportion the A-concentrate by controlling A-concentrate pump **38**. The A-conductivity signal is temperature compensated via a reading from temperature sensor **42**.

[0073] A B-concentrate pump **44**, such as a peristaltic or piston pump, pumps B-concentrate from a B-concentrate source **26** into purified water line **32** via B-concentrate line **36**. Conductivity cell **46** measures the conductive effect of the B-concentrate on the purified water/A-concentrate mixture, sends a signal to logic implementer **20**, which uses the signal to properly proportion the B-concentrate by controlling B-concentrate pump **44**. The B-conductivity signal is also temperature compensated via a reading from temperature sensor **48**.

[0074] A water tank **50** holds purified water prior to receiving the concentrates, which has been degassed in a degassing chamber **51** via a degassing pump **53**, located below water tank **50**. A heater **52** controlled by logic implementer **20** heats the purified water for treatment to body temperature, e.g., 37° C. The fluid exiting conductivity cell **46** is therefore freshly prepared dialysis fluid, properly degassed and heated, and suitable for sending to dialyzer **102** for treatment. A fresh

dialysis fluid pump **54**, such as a gear pump, delivers the fresh dialysis fluid to dialyzer **102**. Logic implementer **20** controls fresh dialysis fluid pump **54** to deliver fresh dialysis fluid to the dialyzer at a specified flowrate as described in more detail below.

[0075] A used dialysis fluid line **56** via a used dialysis fluid pump **58** returns used dialysis fluid from the dialyzer through an external drain line **57** to a drain **60**. Logic implementer **20** controls used dialysis fluid pump **58** to pull used dialysis fluid from dialyzer **102** at a specified flowrate. An air separator **62** separates air from the used dialysis fluid in used dialysis fluid line **56**. A pressure sensor **64** senses the pressure of used dialysis fluid within used dialysis fluid line **56** and sends a corresponding pressure signal to logic implementer **20**.

[0076] Conductivity cell **66** measures the conductivity of used fluid flowing through used dialysis fluid line **56** and sends a signal to logic implementer **20**. The conductivity signal of cell **66** is also temperature compensated via a reading from temperature sensor **68**. A blood leak detector **70**, such as an optical detector, looks for the presence of blood in used dialysis fluid line **56**, e.g., to detect if a dialyzer membrane has a tear or leak. A heat exchanger **72** recoups heat from the used dialysis fluid exiting dialysis fluid circuit **30** to drain **60**, preheating the purified water traveling towards heater **52** to conserve energy.

[0077] A fluid bypass line **74** allows fresh dialysis fluid to flow from fresh dialysis fluid line **76** to used dialysis fluid line **56** without contacting dialyzer **102**. A fresh dialysis fluid tube **78** extends from machine **12** and carries fresh dialysis fluid from fresh dialysis fluid line **76** to dialyzer **102**. A used dialysis fluid tube **80** also extends from machine **12** and carries used dialysis fluid from dialyzer **102** to used dialysis fluid line **56**.

[0078] Fresh dialysis fluid line **76** also includes a conductivity sensor or cell **82** that senses the conductivity of fresh dialysis fluid leaving a UF system **90** and sends a corresponding signal to logic implementer **20**. The conductivity signal of cell **82** is likewise temperature compensated via a reading from temperature sensor **84**.

[0079] An ultrafilter **86** further purifies the fresh dialysis fluid before being delivered via dialysis fluid line **76** and fresh dialysis fluid tube **78** to dialyzer **102**. Alternatively or additionally, one or more ultrafilter (additional ultrafilter is not illustrated) is used to purify the fresh dialysis fluid to the point where it may be used as substitution to perform from pre- or post-dilution hemofiltration or hemodiafiltration.

[0080] UF system **90** monitors the flowrate of fresh dialysis fluid flowing to dialyzer **102** (and/or as substitution fluid flowing directly to the blood set (FIG. 2)) and used fluid flowing from the dialyzer. UF system **90** includes fresh and used flow sensors **Q1c** and **Q2c**, respectively, which send signals to logic implementer **20** indicative of the fresh and used dialysis fluid flowrate, respectively. Logic implementer **20** uses the signals to set used dialysis fluid pump **58** to pump faster than fresh dialysis fluid pump **54** by a predetermined amount to remove a prescribed amount of ultrafiltration (“UF”) from the patient over the course of treatment. Fresh and used flow sensors **Q1p** and **Q2p** are redundant sensors that ensure UF system **90** is functioning properly.

[0081] System **10** provides plural valves **92** (referring collectively to valves **92a** to **92k**) under the control of logic implementer **20** to selectively control a prescribed treatment. In particular, valve **92a** selectively opens and closes bypass line **68**, e.g., to allow disinfection fluid to flow from fresh dialysis fluid line **76** to used dialysis fluid line **56**. Valve **92b** selectively opens and closes fresh dialysis fluid line **76**. Valve **92c** selectively opens and closes used dialysis fluid line **56**. Valve **92d** selectively opens and closes used dialysis fluid line **56** to external drain line **57** and drain **60**. Valve **92e** selectively opens and closes purified water line **32** to purified water source **22**. Valves **92f** and **92g** control A- and B-concentrate flow, respectively. Valves **92h** to **92k** operate with UF system **90**.

[0082] It should be appreciated that the dialysis fluid circuit **30** is simplified and may include other structure (e.g., more valves) and functionality not illustrated. Also, dialysis fluid circuit illustrates a hemodialysis (“HD”) pathway. It is contemplated to provide an additional ultrafilter (not illustrated) in fresh dialysis fluid line **76** to create substitution fluid for hemofiltration (“HF”). It is

also contemplated to provide one or more ultrafilter in one or more line(s) branching off of fresh dialysis fluid line **76** to create substitution fluid, in addition to the fresh dialysis fluid in line **76**, for hemodiafiltration (“HDF”).

[0083] Referring now to FIG. 7, blood circuit or set **100** illustrates one embodiment of a blood set that may be used with either system **10**. Blood circuit or set **100** includes a dialyzer **102** having many hollow fiber semi-permeable membranes **104**, which separate dialyzer **102** into a blood compartment and a dialysis fluid compartment. The dialysis fluid compartment during treatment is placed in fluid communication with a distal end of fresh dialysis fluid tube **78** and a distal end of used dialysis fluid tube **80**. For HF and HDF, a separate substitution tube, in addition to fresh dialysis fluid tube **78**, is placed during treatment in fluid communication with one or both of arterial line **106** extending from an arterial access **106a** and venous line **108** extending to a venous access **108a**. In HDF, dialysis fluid also flows through dialysis fluid tube **78** to dialyzer **102**, while for HF, dialysis fluid flow through tube **78** is blocked.

[0084] An arterial pressure pod **110** may be placed upstream of blood pump **120**, while venous line **108** includes a venous pressure pod **112**. Pressure pods **110** and **112** operate with blood pressure sensors (not illustrated) mounted on the machine housing, which send arterial and venous pressure signals, respectively, to logic implementer **20**. Venous line **108** includes a venous drip chamber **114**, which removes air from the patient's blood before the blood is returned to patient **116**.

[0085] Arterial line **106** of blood circuit or set **100** is operated by blood pump **120**, which is under the control of logic implementer **20** to pump blood at a desired flowrate. System **10** also provides multiple blood side electronic devices that send signals to and/or receive commands from logic implementer **20**. For example, logic implementer **20** commands pinch clamps **122a** and **122b** to selectively open or close arterial line **106** and venous line **108**, respectively. A blood volume sensor (“BVS”) **124** is located along arterial line **106** upstream of blood pump **120**. Air detector **126** looks for air in venous blood line **108**.

[0086] To create the electrically floating dialysis fluid circuit **30**, sensitive equipment, such as conductivity sensors **40**, **46**, **66** and **82**, which are normally connected directly or indirectly to protective earth **27**, are instead left floating with respect to protective earth **27** or other earth within machine **12**. Likewise, the flow sensors **Q1c**, **Q2c**, **Q1p**, and **Q2p** of UF system **90** are not connected to protective earth. In the illustrated embodiment, nowhere within machine **12** is electrically floating dialysis fluid circuit **30** or the floating blood set **100** physically connected to protective earth **27** or other electrical earth. Thus, dialysis fluid circuit **30** and the blood set **100** (including arterial line **106**, venous line **108**, and patient access or needles **106a**, **108a**) with respect to machine **12** are said to be electrically floating.

[0087] The sensitive equipment, however, is in the prior art connected to protective earth **27** for a reason, namely, if not properly earthed, stray current from outside or inside the machine or a faulty component may cause the conductivity and flow sensors to read or output improperly. To combat this problem without connecting the sensors to protective earth **27**, the sensors of the present disclosure are provided with electrical bypass lines **150** as illustrated in FIG. 6. Bypass lines **150** electrically bypass the sensing equipment, from a point upstream of the sensors to a point downstream of the sensors, so that fault currents conduct from upstream to downstream of each sensor, or vice versa, through electrical bypasses **150** and not the sensing equipment. In doing so, bypass lines **150** electrically contact, or are otherwise in electrical communication with, fluid upstream of the sensors and fluid downstream of the sensors, creating a short circuit around the sensors. The short circuit causes stray currents to bypass the sensors, so that the stray currents do not affect the operation of the sensors.

[0088] Additionally, the sensors and any other flow component conductively touching liquid in the dialysis fluid circuit **30** and the blood set **100** is electrically insulated from the remainder of machine **12** via mechanical insulation. Mechanical insulation refers to the use of a non-conductive material, e.g., plastic, rubber, ceramic, and combinations thereof, placed between the fluid

contacting component and the machine. The result may be an insulating pad located between the component and the machine chassis or other machine fixture to which the component is mounted. [0089] Besides, the “mechanical” electrical insulation, to make the dialysis fluid circuit **30** and the blood set **100** floating, electrical power wires and electrical signal wires that conduct signals away from the sensor probes or other structures that contact the blood, dialysis fluid or concentrate, for example, need to be electrically isolated from the circuitry and computational devices that read and analyze the sensor signals, e.g., printed circuit boards, processing, memory (discussed above as logic implementer **20**). To do so, each power and signal wire stemming from a sensor is in one embodiment isolated via a transformer (or any of the described above for electrically insulated sockets **170a** to **170c**) from a power or signal wire, respectively, that extends from the transformer or other electrically insulating device to logic implementer **20** or power source. The transformers or other electrically insulating devices pass along the information carried by the sensor signal wiring, while creating a physical break in the signal lines. The physical breaks prevent (i) stray currents from machine **12** from entering the electrically floating fluid pathway **140** via the sensor power or signal lines and (ii) stray currents within floating fluid pathway **140** from exiting out to machine **12** and its other components via the sensor power or signal lines.

[0090] So to make the dialysis fluid circuit **30** and the blood set **100** electrically floating within machine **12**, four features may be ensured: (i) no connection from a conductive fluid path to protective earth **27** or other electrical earth is made within machine **12** or in the blood set **100**, (ii) sensitive equipment that touches conductive water, concentrate, dialysis fluid and/or blood in dialysis fluid circuit **30** and blood in blood set **100** is electrically bypassed, (iii) components that contact liquid are “mechanically” electrically insulated when mounted, and (iv) signal wires to or from logic implementer **20** are electrically isolated.

[0091] The above-described combination of different electrical insulation provides a significantly safer renal failure therapy system **10** versus known systems. It should be appreciated that the insulated electrical sockets of the present disclosure do not have to be used with the electrically floating fluid pathway **140**, and vice versa, and that each make system **10** safer even if used alone. The present disclosure therefore does not require such combination.

ELEMENT NUMBER LISTING

[0092] known dialysis machine **8** [0093] renal failure therapy system **10** [0094] renal failure therapy machine **12** [0095] user interface **14**, [0096] at least one processor **16**, [0097] at least one memory **18**, [0098] logic implementer **20**, [0099] purified water device or source **22**, [0100] A-concentrate source **24**, [0101] B-concentrate source **26**, [0102] machine protective earth **27**, [0103] electrical earth **28**, [0104] dialysis fluid circuit **30**, [0105] purified water line **32**, [0106] A-concentrate line **34**, [0107] B-concentrate line **36**, [0108] A-concentrate pump **38**, [0109] conductivity cell **40**, [0110] temperature sensor **42**, [0111] B-concentrate pump **44**, [0112] conductivity cell **46**, [0113] temperature sensor **48**, [0114] water tank **50**, [0115] degassing chamber **51**, [0116] dialysis fluid heater **52**, [0117] degassing pump **53**, [0118] fresh dialysis fluid pump **54**, [0119] used dialysis fluid line **56**, [0120] external drain line **57**, [0121] used dialysis fluid pump **58**, [0122] drain **60**, [0123] air separator **62**, [0124] pressure sensor **64**, [0125] dialyzer **102**, [0126] conductivity cell **66**, [0127] temperature sensor **68**, [0128] blood leak detector **70**, [0129] heat exchanger **72**, [0130] fluid bypass line **74**, [0131] fresh dialysis fluid line **76**, [0132] fresh dialysis fluid tube **78**, [0133] used dialysis fluid tube **80**, [0134] conductivity cell **82**, [0135] temperature sensor **84**, [0136] ultrafilter **86**, [0137] UF system **90**, [0138] plural valves **92** (collectively to valves **92a** to **92k**), [0139] blood circuit or set **100**, [0140] dialyzer **102**, [0141] semi-permeable membranes **104**, [0142] arterial line **106**, [0143] arterial access or needle **106a**, [0144] venous line **108**, [0145] venous access or needle **108a**, [0146] arterial pressure pod **110**, [0147] venous pressure pod **112**, [0148] venous drip chamber **114**, [0149] patient **116**, [0150] blood pump **120**, [0151] pinch clamps **122a** and **122b**, [0152] blood volume sensor (“BVS”) **124**, [0153] air detector **126**, [0154] floating fluid pathway **140**, [0155] electrical bypass lines **150**, [0156]

mains transformer **152**, [0157] mains input coil **154**, [0158] mains output coil **156**, [0159] isolation barrier **158**, [0160] location **160** for electrically insulated sockets **170**, [0161] banks **162**, **164**, and **166** of electrically insulated sockets **170**, [0162] electrically insulated electrical sockets **170** (collectively to sockets **170a** to **170c**), [0163] AC to AC transformer **172**, [0164] primary coil **174**, [0165] secondary coil **176**, [0166] isolation barrier **178**, [0167] DC to DC converter **180** having input **180a** and output **180b**, [0168] power wire **182**, [0169] power ground wires **184**, [0170] AC to DC conversion **192**, [0171] DC to AC conversion **194** [0172] external electrical device **200** [0173] It should be understood that various changes and modifications to the presently preferred embodiments described herein will be apparent to those skilled in the art. Such changes and modifications may be made without departing from the spirit and scope of the present invention and without diminishing its intended advantages. It is therefore intended that such changes and modifications be covered by the appended claims.

Claims

1. A method for powering an external device through a renal failure therapy system, the renal failure therapy system comprising: a dialyzer; a blood circuit in fluid communication with the dialyzer; a dialysis fluid circuit in fluid communication with the dialyzer; a housing supporting the dialyzer, the blood circuit, and the dialysis fluid circuit; and at least one electrical socket held by the housing, wherein the method comprises providing, through the at least one electrical socket, a voltage output to a respective external electrical device for powering or charging the respective external electrical device, and wherein the at least one electrical socket includes electrical insulation for protecting a patient while powering or charging the respective external electrical device.
2. The method of claim 1, wherein the at least one electrical socket is a Universal Serial Bus (USB) socket or a 120 VAC or 220 VAC socket.
3. The method of claim 1, wherein the voltage output is selected from the group consisting of a 120 VAC output and a 240 VAC output.
4. The method of claim 1, wherein the electrical insulation is configured to provide additional galvanic separation between the patient and the renal failure therapy system to minimize leakage or fault currents.
5. The method of claim 1, further comprising: providing a transformer in the at least one electrical socket to obtain the electrical insulation, the transformer providing galvanic isolation, forming an open circuit, and preventing propagation of fault currents, the transformer having an input coil separated physically from an output coil, the input coil magnetically inducing an AC output voltage in the output coil, the AC output voltage powering or charging the respective external electrical device; and providing a mains transformer in the renal failure therapy system, the mains transformer being a second layer of electrical insulation between the patient and mains power, and comprising a mains input coil and a mains output coil separated by an isolation barrier, the mains transformer being located between mains power and input to the electrical socket, the isolation barrier preventing conducting of electricity, while mains input coil induces a desired voltage in mains output coil.
6. The method of claim 1, further comprising providing a transformer in the at least one electrical socket to obtain the electrical insulation, the transformer providing galvanic isolation forming an open circuit, which prevents propagation of fault currents, the transformer having an input coil separated physically from an output coil, the input coil magnetically inducing a desired AC output voltage in the output coil, the desired AC output voltage charging or powering the respective external electrical device.
7. The method of claim 1, wherein the electrical insulation of the at least one electrical socket includes a DC-to-DC converter.

8. The method of claim 7, wherein the DC-to-DC converter includes an input conductively insulated from an output.
9. The method of claim 1, wherein the electrical insulation of the at least one electrical socket includes a transformer in combination with a DC-to-AC converter.
10. The method of claim 1, wherein the electrical insulation includes double electrical insulation.
11. The method of claim 1, wherein the at least one electrical socket includes: (i) a first electrical socket and the voltage output is a first voltage output, and (ii) a second electrical socket configured to provide a second voltage output different than the first voltage output.
12. The method of claim 1, wherein the at least one electrical socket includes (i) an AC current socket and the voltage output is an AC voltage output, and (ii) a DC current socket configured to provide a DC voltage output different than the AC voltage output, the method further comprising: electrically insulating the AC current socket via a transformer that provides galvanic isolation forming an open circuit, which prevents propagation of fault currents, the transformer having an input coil separated physically from an output coil, magnetically inducing via the input coil a desired AC output voltage in the output coil and the desired AC output voltage charging or powering the respective external electrical device, electrically insulating the DC current socket, the DC current socket comprising a DC to AC converter, an auxiliary transformer and an isolation barrier conductively isolating the input AC voltage from an AC output voltage, inducing via a primary coil the AC output voltage in a secondary coil, an AC to DC converter being provided on an output side of the auxiliary transformer, converting an input DC voltage to an input AC voltage, inducing the AC output voltage in secondary coil, and converting the induced AC output voltage to a desired DC output.
13. The method of claim 1, wherein the at least one electrical socket includes a bank of electrical sockets configured to provide voltage outputs for respective external electrical devices.
14. The method of claim 13, wherein the bank is a first bank, the renal failure therapy system further comprising a second bank of electrical sockets configured to provide a second voltage output dedicated to a voltage of a second external electrical device, the second voltage output being different from the first voltage output.
15. The method of claim 1, wherein the at least one electrical socket is located on a front, side, top, or backside of the housing.
16. A method for powering an external device through a system, the system comprising: an external electrical device; and a renal failure therapy system including: a dialyzer, a blood circuit in fluid communication with the dialyzer, a dialysis fluid circuit in fluid communication with the dialyzer, a housing supporting the dialyzer, the blood circuit and the dialysis fluid circuit, and at least one electrical socket held by the housing, the at least one electrical socket including a transformer, wherein the method comprises: providing, through the at least one electrical socket, a voltage output to the external electrical device for powering or charging the external electrical device via the transformer, providing, through the transformer, a galvanic isolation to form an open circuit, and preventing propagation of fault currents, the transformer having an input coil separated physically from an output coil, and magnetically inducing, through the input coil, an AC output voltage in the output coil and the AC output voltage powering or charging the external electrical device, wherein the at least one electrical socket includes electrical insulation for protecting a patient while powering or charging the external electrical device.
17. The method of claim 16, wherein the external electrical device is a personal communication/computing device, a reading lamp, or a heating pad.
18. The method of claim 16, wherein the at least one electrical socket is a Universal Serial Bus (USB) socket or a 120 VAC or 220 VAC socket.
19. The method of claim 16, wherein the voltage output is selected from the group consisting of a 120 VAC output and a 240 VAC output.
20. A method for powering an external device through a renal failure therapy system, the renal

failure therapy system comprising: a dialyzer; a blood circuit in fluid communication with the dialyzer; a dialysis fluid circuit in fluid communication with the dialyzer; a housing supporting the dialyzer, the blood circuit and the dialysis fluid circuit; and at least one electrical socket being an AC current socket and being held by the housing, wherein the method comprises providing, through the at least one electrical socket, a voltage output to a respective external electrical device for powering or charging the respective external electrical device, wherein the method comprises: providing electrical insulation in the at least one electrical socket for protecting a patient while powering or charging the respective external electrical device, wherein the electrical insulation of the at least one electrical socket includes a transformer that provides galvanic isolation forming an open circuit, which prevents propagation of fault currents, the transformer having an input coil separated physically from an output coil, magnetically inducing, through the input coil, a desired AC output voltage in the output coil, the desired AC output voltage charging or powering the respective external electrical device, providing a mains transformer, being a second layer of electrical insulation between the patient and mains power, the mains transformer having a mains input coil and a mains output coil separated by an isolation barrier, the mains transformer being located between mains power and input or inputs to the at least one electrical socket, and preventing, through the isolation barrier, a conduction of electricity, while mains input coil induces a desired voltage in mains output coil.
